StudyFlow Pro

Product Requirements Document

Study Tracker + Habit Manager + AI Planner

Version	Date	Status	Platform
1.0	Feb 2026	Draft	Web (Next.js 14)

Technology Stack

Layer	Technology	Purpose	License
Frontend	Next.js 14 + React	SSR/SSG framework	MIT (Free)
Styling	Tailwind CSS + shadcn/ui	Utility-first + components	MIT (Free)
Animations	GSAP + Framer Motion	Scroll + micro-interactions	Free/GSAP Free Tier
Icons	Lucide React + Lottie	Static + animated icons	MIT / Free Tier
Auth	NextAuth.js + Google OAuth	Secure authentication	MIT (Free)
Database	Supabase (PostgreSQL)	Data storage + real-time	Free tier 500MB
ORM	Drizzle ORM	Type-safe DB queries	MIT (Free)
Charts	Recharts	Data visualization	MIT (Free)
State	Zustand	Global state management	MIT (Free)
AI	Claude API / OpenAI API	Smart plan generation	Pay-per-use
Deploy	Vercel	Hosting + edge network	Free hobby tier

1. Product Overview

1.1 Vision

StudyFlow Pro is an all-in-one productivity platform that combines AI-assisted exam and study planning, time-bounded topic scheduling, open-ended task estimation, daily habit tracking, streak-based motivation, and rich analytics — all wrapped in a smooth, dark-themed, animated interface.

Core Value Proposition

Think Notion + AI Planner + Habit Tracker: Users define their goals (exam date, work deadlines, skill targets), and the platform automatically distributes topics across their available time, tracks daily progress, and gamifies consistency through streaks and visual charts — inspired by the reference UIs from Routiner, Everyday, HabitKit, and Tasker shown in the design references.

1.2 Target Users

- Students preparing for competitive exams (UPSC, GATE, JEE, NEET, GRE, IELTS)
- Professionals tracking work projects with deadlines
- Self-learners working through courses or skill acquisition
- Anyone building consistent daily/weekly habits and routines

1.3 Key Differentiators

- AI-generated smart study plans from exam name or uploaded syllabus
- Dual planning modes: date-bounded (exam prep) and open-ended (daily hours → completion estimate)
- Habit tracker integrated in the same app with cross-module streaks
- Animated, gamified streak system with color evolution (red → orange → gold → purple → fire)
- Fully dark-themed with tasteful GSAP + Framer Motion animations only where they add value

2. Site Map & Navigation

Route	Page Name	Access	Description
/	Landing Page	Public	Hero, features, CTA, animated showcase
/login	Auth Page	Public	Google OAuth sign-in via NextAuth
/dashboard	Dashboard	Private	Overview: streaks, today's tasks, habit rings, quick stats
/planner	Study Planner	Private	Create/manage study/work/exam plans
/planner/new	New Plan	Private	AI-assisted or manual plan creation flow
/planner/[id]	Plan Detail	Private	Topic list, daily schedule, progress tracking
/habits	Habit Tracker	Private	Daily/weekly/monthly habit management
/habits/progress	Habit Analytics	Private	Charts, graphs, streaks, heatmaps
/analytics	Analytics Hub	Private	Cross-module progress reports and insights
/settings	Settings	Private	Profile, notifications, preferences

2.1 Navigation Layout

Persistent sidebar (desktop) / bottom tab bar (mobile) with animated icons using Lucide React. Active states use GSAP scale + color transitions. The sidebar collapses to icon-only mode with smooth Framer Motion width animation.

- Sidebar sections: Dashboard, Planner, Habits, Analytics, Settings
- Top bar: User avatar, streak badge (animated fire icon), global search, notifications
- Mobile: bottom tab navigation with spring animations on tab switch

3. Authentication & User Management

3.1 Authentication Flow

Authentication is handled entirely by NextAuth.js with Google OAuth 2.0. No username/password — Google login only for simplicity and security.

Step	Action	Technical Detail
1	User clicks 'Sign in with Google'	NextAuth triggers Google OAuth consent screen
2	Google returns access token	NextAuth creates JWT session (httpOnly cookie)
3	New user check	Supabase upserts user record on first login
4	Redirect to /dashboard	Session available via useSession() hook
5	Subsequent visits	JWT auto-refresh, no re-login needed

3.2 Security

- All API routes protected with getServerSession() middleware
- User data is row-level secured in Supabase (RLS policies — users can only read/write their own rows)
- No sensitive data in localStorage — session managed server-side via NextAuth JWT
- CSRF protection built into NextAuth
- Environment variables: GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, NEXTAUTH_SECRET stored in Vercel env
- Supabase service role key never exposed to client — all DB ops through server-side API routes

4. Database Schema (Supabase / PostgreSQL)

4.1 Core Tables

users

Column	Type	Description
id	UUID PK	Google OAuth sub ID
email	TEXT UNIQUE	User email from Google
name	TEXT	Display name
avatar_url	TEXT	Google profile picture URL
created_at	TIMESTAMPTZ	Account creation time
settings	JSONB	User preferences (theme, notifications, timezone)

plans

Column	Type	Description
id	UUID PK	Plan unique ID
user_id	UUID FK	References users.id
title	TEXT	Plan name (e.g. 'GATE 2026 Prep')
type	ENUM	'exam' | 'work' | 'course' | 'custom'
start_date	DATE	Plan start date
end_date	DATE NULLABLE	NULL = open-ended plan
daily_hours	FLOAT NULLABLE	Hours/day for open-ended plans
total_estimated_hours	FLOAT	Sum of all topic hours (computed)
color	TEXT	Hex color for UI differentiation
is_ai_generated	BOOLEAN	Was AI used to create this plan
status	ENUM	'active' | 'completed' | 'paused'
created_at	TIMESTAMPTZ	Creation timestamp

topics

Column	Type	Description
id	UUID PK	Topic unique ID
plan_id	UUID FK	References plans.id
parent_id	UUID FK NULLABLE	NULL = top-level topic; set for subtopics
title	TEXT	Topic name
estimated_hours	FLOAT	Hours to complete this topic
priority	INTEGER	1 (highest) to 5 (lowest)
weightage	FLOAT NULLABLE	Exam weightage percentage (optional)
scheduled_date	DATE NULLABLE	Auto-assigned date from planner algorithm
order_index	INTEGER	Sort order within parent
status	ENUM	'not_started' | 'in_progress' | 'completed'
notes	TEXT	User notes for this topic

daily_progress

Column	Type	Description
id	UUID PK	Entry unique ID
user_id	UUID FK	References users.id
topic_id	UUID FK	References topics.id
date	DATE	The date this study session occurred
hours_spent	FLOAT	Actual hours studied
completion_pct	INTEGER	0-100 percent complete
notes	TEXT	Session notes
created_at	TIMESTAMPTZ	Entry created timestamp

habits

Column	Type	Description
id	UUID PK	Habit unique ID
user_id	UUID FK	References users.id
title	TEXT	Habit name (e.g. 'Meditate 10 min')
category	TEXT	Health, Mind, Study, Work, etc.

Column	Type	Description
frequency	ENUM	'daily' | 'weekly' | 'monthly'
target_days	INT[]	Days of week [0-6] for weekly habits
color	TEXT	Hex color for habit
icon	TEXT	Lucide icon name
is_active	BOOLEAN	Soft delete / pause
created_at	TIMESTAMPTZ	Creation timestamp

habit_logs

Column	Type	Description
id	UUID PK	Log entry ID
habit_id	UUID FK	References habits.id
user_id	UUID FK	References users.id
date	DATE	Log date
status	ENUM	'done' | 'missed' | 'skipped'
note	TEXT NULLABLE	Optional note

streaks

Column	Type	Description
id	UUID PK	Streak record ID
user_id	UUID FK	References users.id
type	ENUM	'global' | 'plan' | 'habit'
ref_id	UUID NULLABLE	plan_id or habit_id if type is not global
current_streak	INTEGER	Current consecutive days
longest_streak	INTEGER	All-time best streak
last_active_date	DATE	Last date user was active
updated_at	TIMESTAMPTZ	Last update time

5. Feature Specifications

5.1 Dashboard (/ dashboard)

The main hub users see after login. Inspired by the Virtus dark dashboard (Image 2) and Tasker (Image 10) reference designs.

5.1.1 Dashboard Components

- Streak Banner — animated fire icon (Lottie) showing current global streak. Color evolves: grey (0) → red (1-2) → orange (3-6) → gold (7-13) → purple (14-29) → animated rainbow fire (30+). Missing a day resets to 0 with a shake animation.
- Today's Focus Card — top priority topics due today with one-click completion checkboxes. GSAP stagger animation on load.
- Active Plans Summary — horizontal scrollable card row showing each plan's progress ring (Framer Motion animated on mount).
- Habit Rings — circular completion rings for today's habits (inspired by Apple Rings). Animated SVG stroke-dashoffset with Framer Motion.
- Quick Stats Row — cards for: Study hours this week, Habits completed today, Days till next exam, Longest streak. CountUp animation with GSAP.
- Recent Activity Feed — last 5 study/habit log entries with timestamps.
- Motivational Quote — changes daily, fades in with Framer Motion opacity animation.

5.2 Study Planner (/planner)

5.2.1 Plan List View

- Card grid of all active/past plans with progress bar, date range, type badge, and color swatch
- '+ New Plan' button with Framer Motion scale animation on hover
- Filter/sort by: type, status, date, completion %
- Quick stats bar: total active plans, total hours tracked, overall completion rate

5.2.2 New Plan Creation (/planner/new)

Two creation modes available: Manual and AI-Assisted.

Manual Plan Creation Flow

Step 1: Enter plan title, type (exam/work/course/custom), colorStep 2: Choose mode — Date-bounded (select start + end date) or Open-ended (enter daily available hours)Step 3: Add topics manually — topic name, estimated hours, priority (1-5), optional weightage %Step 4: Add subtopics under any topic (nested, max 2 levels)Step 5: Review auto-scheduled plan → Confirm

AI-Assisted Plan Creation Flow (⭐ Key Feature)

Step 1: Enter exam/goal name (e.g. 'GATE CS 2026') OR type custom goal text OR upload syllabus PDFStep 2: Click ✨ 'Generate Smart Plan'Step 3: AI returns: topics, subtopics, estimated hours per topic, suggested priority, suggested daily hoursStep 4: Editable table — user can modify any field (add/remove/edit topics, change hours, reorder)Step 5: User clicks Confirm → Plan created with full scheduleAPI: Claude API (claude-sonnet-4-6) or OpenAI GPT-4o with structured JSON output. Prompt includes exam name, date range, and instructions to return topics as a JSON array.

5.2.3 Plan Scheduling Algorithm

When a plan is confirmed, the backend scheduling algorithm runs:

- Date-bounded mode: Total hours = sum of all topic hours. Available days = end_date - start_date (excluding user-marked off days). Daily allocation = total_hours / available_days. Topics assigned to dates priority-first (highest priority → earliest dates). Subtopics inherit parent's scheduled date or span adjacent days if hours exceed daily limit.
- Open-ended mode: User enters daily available hours. Algorithm computes: days_needed = ceil(total_hours / daily_hours). Shows a completion estimate date (e.g. 'You will finish in 34 days — by April 5th'). No fixed end date; dynamically updates as user logs progress.
- Re-scheduling: If user falls behind (misses a day or marks less progress than planned), a 'Reschedule' button appears. Algorithm redistributes remaining topics across remaining days.

5.2.4 Plan Detail View (/planner/[id])

- Header: Plan title, type badge, date range or estimated completion, overall progress ring
- Timeline View: Calendar-like grid (inspired by Image 12 — dark calendar dashboard) showing which topics are scheduled on which days. Colored blocks per topic.
- Topic List: Expandable tree showing topics → subtopics. Each row has: title, estimated hours, actual hours logged, priority badge, status selector (not started / in progress / completed), progress bar.
- Daily Log Panel: Right sidebar — select a date, pick a topic/subtopic, log hours spent + completion %. Framer Motion slide-in animation.
- Progress Charts: Line chart (hours planned vs actual), bar chart (daily completion), area chart (cumulative progress). Recharts with custom dark theme.
- Insights Panel: 'At this pace you will finish on [date]', 'You are 3 days behind schedule', 'Top remaining topic: [X]'.

5.3 Habit Tracker (/habits)

Inspired by: Routiner (Image 1), Everyday heatmap (Image 3), HabitKit dot grid (Image 4/5), HabitBull (Image 6), Habit tracker app (Images 7/9), and the Notion habit tracker (Image 8).

5.3.1 Habit List / Daily View

- Today view: List of all active habits for today. Each row shows habit name, category icon, streak badge, yes/no/skip action buttons (inspired by Images 7/9).

- Week strip at top: Mon-Sun with dots marking completion (green = done, red = missed, gray = future/skipped).
- Category filter tabs: All, Health, Mind, Study, Work, Custom.
- '+ Add Habit' button with modal form: name, category, frequency (daily/weekly/monthly), target days, color, icon.
- Animated completion: Checking a habit triggers a Lottie confetti burst + streak badge increment animation.

5.3.2 Habit Progress Page (/habits/progress)

- Heatmap Calendar (like Everyday — Image 3): Full year or 6-month heatmap grid for each habit. Color intensity = completion intensity. Built with custom React + Recharts or D3.
- Dot Grid View (like HabitKit — Images 4/5): A grid of colored squares representing each day — each dot is a completed day. Hover tooltip shows date + note.
- Streaks Panel: Current streak, longest streak, start date of current streak. Streak history timeline bar chart (like Image 4 Streaks widget).
- Bar Chart: Weekly/monthly completion rate bars per habit. Toggle between habit views.
- Pie Chart: Overall completion breakdown — Done / Missed / Skipped (like Image 7 pie chart).
- Completion Rate Card: Large percentage display with trend arrow (like Image 4/5 — '69.5% Avg Completion Rate').
- Time range selector: 2w / 12w / 3m / 6m / 1y / 3y (like Image 7).

6. Streak System

The streak system is the core engagement mechanic. It is designed to be addictive but fair, inspired by Duolingo and Snapchat streak models.

Streak Days	Title	Badge Color	Animation
0	No Streak	Gray (#6B7280)	Static dash icon
1–2	Getting Started	Red (#EF4444)	Slow pulse
3–6	On a Roll	Orange (#F97316)	Medium pulse + glow
7–13	One Week Strong	Yellow (#EAB308)	Fast pulse + glow ring
14–29	Two Weeks Warrior	Purple (#8B5CF6)	Particle orbit
30–59	Month Master	Cyan (#06B6D4)	Flame animation (Lottie)
60–99	Unstoppable	Gold gradient	Rainbow flame Lottie
100+	Legend	Animated rainbow	Explosive rainbow Lottie

6.1 Streak Logic Rules

- Global streak: Increments if user logs ANY progress (study OR habit completion) on a given day
- Habit streak: Per-habit streak increments only when that specific habit is marked 'done' for the day
- Plan streak: Increments when user logs any study time for a plan on the given day
- Missed day: If user does not log any activity by end of day (midnight in user's timezone), streak resets to 0
- Skip: Marking a habit as 'skipped' does NOT break the streak (grace mechanic)
- Streak freeze: Future feature — allow 1 freeze per 7-day period (configurable in settings)
- Reset animation: When streak breaks, a dramatic 'shatter' Framer Motion animation plays on the badge
- Recovery: After break, a subtle 'Start Again' motivational message with CTA to log today's progress

7. AI-Assisted Planning Module

7.1 Overview

The AI planner bridges the gap between a user's goal ('I need to prepare for GATE CS by December') and a structured, time-distributed study plan. It is not fully automatic — it generates a suggestion that the user reviews and edits before confirming. This design mirrors Notion AI's approach: AI as a starting point, human as the decision-maker.

7.2 AI Input Options

Input Type	Example	Processing
Exam name	'GATE CS 2026'	AI has pre-trained knowledge of standard syllabus structure
Custom goal	'Learn React in 2 months'	AI extracts skill topics from goal description
Syllabus text	Paste text from PDF	AI parses and structures into topics/subtopics
PDF upload	Upload syllabus PDF	Extract text via pdf-parse, then send to AI

7.3 AI Output Schema

The AI returns a structured JSON array:

```
{ "topics": [{ "title": "Data Structures", "estimated_hours": 20, "priority": 1, "weightage": 15, "subtopics": [{ "title": "Arrays & Strings", "estimated_hours": 4, "priority": 1 }, { "title": "Linked Lists", "estimated_hours": 5, "priority": 1 }] }] }
```

7.4 Editable Review Table

After AI generation, results appear in a fully interactive table with these editable columns: Topic name, Estimated hours (number input), Priority (1-5 star selector), Weightage %, Add/Remove subtopics, Reorder (drag handle). User can also add new rows manually. Table has live total hours counter and estimated completion date shown in real-time as values change.

7.5 API Integration

- Endpoint: POST /api/ai/generate-plan
- Auth: Requires valid NextAuth session (server-side check)
- AI call happens server-side (never expose API key to client)
- Rate limiting: max 5 AI plan generations per user per day (to control costs)
- Fallback: If AI API is unavailable, show manual creation form with example pre-filled data
- Cost optimization: Cache common exam plans (GATE, GRE, etc.) in Supabase so the same AI generation is not repeated

8. Animation Specification

Animations should enhance UX, not distract. Every animation has a purpose. The guiding principle: 'Animate to inform, not to impress.'

Element	Animation Type	Library	Trigger	Duration
Page transition	Fade + slide Y(20px→0)	Framer Motion	Route change	300ms
Sidebar nav items	Scale 1→1.05 + color shift	Framer Motion	Hover	150ms
Sidebar collapse	Width 240→64px	Framer Motion	Click toggle	300ms spring
Dashboard stat cards	CountUp number animation	GSAP	Mount / scroll into view	1200ms
Progress rings	SVG stroke-dashoffset	Framer Motion	Mount / value change	800ms ease
Streak badge	Pulse + scale + color	Framer Motion	Streak increment	400ms spring
Streak break	Shatter / shake	Framer Motion	Missed day detected	600ms
Habit check	Lottie confetti burst	Lottie	Mark as done	Instant
Button hover	Scale 1→1.03 + shadow lift	Framer Motion	Hover	120ms
Button click	Scale 1→0.97 (press)	Framer Motion	Click	80ms
Card hover	translateY(-2px) + shadow	CSS + Framer Motion	Hover	200ms
Topic completion	Strikethrough + fade 50%	Framer Motion	Status = complete	300ms
Scroll text reveal	Opacity 0→1 + Y 30→0	GSAP ScrollTrigger	Scroll into view	600ms stagger
Modal open/close	Scale 0.9→1 + fade	Framer Motion	Toggle	250ms spring
Heatmap cells	Scale 0→1 stagger	Framer Motion	Page mount	20ms per cell
Chart bars/lines	Draw from 0	Recharts native	Mount	800ms ease-out
Lottie icons nav	Play on hover	Lottie React	Hover	Loop while hovered

Element	Animation Type	Library	Trigger	Duration
AI generating	Shimmer skeleton	CSS keyframe	During AI call	Loop until resolved
Notification badge	Bounce	Framer Motion	New notification	300ms spring

Animation Guidelines

✓ Prefer spring physics over linear easing (feels natural)✓ Use layout animations (Framer Motion layoutId) for shared element transitions✓ Respect prefers-reduced-motion: all animations disabled when user setting is on✗ Never animate on every scroll pixel — use ScrollTrigger with threshold✗ No auto-playing videos or looping animations that distract in data tables✗ Limit Lottie to max 3 active instances at once (performance)

9. Design System

9.1 Color Palette

Token	Hex	Usage
bg-base	#0F0F1A	Main page background
bg-surface	#1A1A2E	Cards, panels, modals
bg-elevated	#252540	Hover states, inputs, tooltips
border	#2D2D4E	Card borders, dividers
text-primary	#F9FAFB	Headings, important text
text-secondary	#9CA3AF	Body text, descriptions
text-muted	#6B7280	Placeholders, timestamps
accent-purple	#7C3AED	Primary brand color, CTAs
accent-cyan	#06B6D4	Secondary accent, links
accent-orange	#F97316	Warnings, streak low
accent-green	#10B981	Success, completion
accent-red	#EF4444	Error, missed, failed
streak-gold	#F59E0B	Streak milestone color

9.2 Typography

Element	Font	Size	Weight	Usage
Display	Inter / Geist	48-72px	700-900	Hero headlines only
H1	Inter	32px	700	Page titles
H2	Inter	24px	600	Section headers
H3	Inter	18px	600	Card titles
Body	Inter	14-16px	400	Main content
Caption	Inter	12px	400	Timestamps, hints
Mono	JetBrains Mono	13px	400	Hours, numbers, code

9.3 Component Library (shadcn/ui base + custom)

- Button — variants: primary (purple gradient), secondary (surface), ghost, destructive. All with Framer Motion hover/press animation.
- Card — dark surface, subtle border, hover lift effect. Variants: stat card, plan card, habit card.

- Input / Textarea — dark bg, purple focus ring with glow effect.
- Badge — color-coded for priority (P1=red, P2=orange, P3=yellow, P4=blue, P5=gray), status, category.
- Progress Bar — animated fill, gradient from accent-purple to accent-cyan.
- Progress Ring — SVG-based circular progress with animated stroke.
- Modal / Sheet — Framer Motion scale + fade animation. Dark backdrop blur.
- Tooltip — Framer Motion fade in on hover, shows within 300ms delay.
- Skeleton — shimmer animation for loading states (CSS @keyframes).
- Tabs — underline indicator with Framer Motion layout animation.
- DatePicker — custom built or react-day-picker with dark theme override.

10. API Routes

Method	Route	Auth	Description
GET	/api/user/profile	✓	Get current user profile + settings
PATCH	/api/user/profile	✓	Update name, avatar, settings
GET	/api/plans	✓	List all plans for current user
POST	/api/plans	✓	Create a new plan (manual)
GET	/api/plans/[id]	✓	Get plan with topics + schedule
PATCH	/api/plans/[id]	✓	Update plan details
DELETE	/api/plans/[id]	✓	Soft-delete a plan
POST	/api/plans/[id]/reschedule	✓	Re-run scheduling algorithm
GET	/api/topics?plan_id=	✓	Get all topics for a plan
POST	/api/topics	✓	Create a topic or subtopic
PATCH	/api/topics/[id]	✓	Update topic (status, hours, priority)
DELETE	/api/topics/[id]	✓	Delete topic and its subtopics
POST	/api/progress	✓	Log study progress for a topic
GET	/api/progress?plan_id=&date=	✓	Get progress entries (filterable)
GET	/api/habits	✓	List all habits
POST	/api/habits	✓	Create a new habit
PATCH	/api/habits/[id]	✓	Update habit
DELETE	/api/habits/[id]	✓	Deactivate habit
POST	/api/habits/log	✓	Log daily habit completion
GET	/api/habits/logs?habit_id=&from=&to=	✓	Get habit logs for range
GET	/api/streaks	✓	Get all streaks for user
POST	/api/streaks/check	✓	Recalculate streaks (cron job)
GET	/api/analytics/summary	✓	Aggregate stats for dashboard

Method	Route	Auth	Description
POST	/api/ai/generate-plan	✓	AI-assisted plan generation

11. Analytics Module (/analytics)

11.1 Study Analytics

- Hours studied per day — area chart, last 30/60/90 days
- Hours planned vs actual — dual-line chart, shows deviation
- Completion rate per plan — horizontal bar chart
- Topic breakdown — donut chart of time spent per topic category
- Best study day/time — heatmap grid (hour of day × day of week)
- Pace analysis — 'At current pace, you will complete X plan by [date]'

11.2 Habit Analytics

- Overall habit score — large % display with weekly trend arrow
- Per-habit completion rate — bar chart, color-coded by habit
- Habit heatmap — GitHub contribution-style grid (inspired by Image 3)
- Streak leaderboard — personal ranking of habits by longest streak
- Monthly calendar view — showing habit completion by day

11.3 Overview Stats

- Total study hours (all time, this month, this week)
- Active plans count with % completion distribution
- Global streak — current and all-time best
- Habits completed today vs total active habits
- Upcoming deadlines in the next 7 days

12. Notifications & Reminders

Notification Type	Trigger	Channel	Configurable
Daily study reminder	Time set by user (default 9am)	Browser push	Yes — time + on/off
Habit reminder	Time set per habit	Browser push	Yes — per habit
Streak at risk	6pm if no activity logged that day	Browser push + in-app	Yes
Streak milestone	Streak reaches 7, 14, 30, 60, 100	In-app celebration modal	No
Plan behind schedule	When actual < planned by >15%	In-app banner	Yes
Exam approaching	7 days, 3 days, 1 day before end date	In-app banner + push	Yes
Topic completion	When topic marked complete	In-app toast	No
AI plan ready	When AI generation completes	In-app toast	No

Implementation: Notifications use the Web Push API with service worker. Browser permission is requested after first login. In-app notifications stored in a notifications table in Supabase. A Vercel Cron Job runs every hour to check streak risk and plan deviation.

13. Performance & Technical Requirements

Metric	Target	Strategy
First Contentful Paint	< 1.2s	Static landing page, Next.js SSG
Time to Interactive	< 2.5s	Route-based code splitting, lazy imports
Lighthouse Score	> 90 (all)	Image optimization, font preload, CSP headers
API Response Time	< 300ms p95	Supabase connection pooling, edge functions
Animation FPS	60fps consistent	will-change CSS, GPU-accelerated transforms only
Bundle Size	< 150KB initial JS	Tree shaking, dynamic imports for charts/Lottie
Supabase Queries	< 50ms p95	Indexed columns: user_id, date, plan_id, habit_id

13.1 Supabase Row Level Security (RLS) Policies

Every table has RLS enabled. Example policy for plans table:

```
CREATE POLICY user_plans ON plans USING (user_id = auth.uid());
```

Same pattern for topics, daily_progress, habits, habit_logs, streaks. This ensures users can only ever access their own data, even if a bug exists in the application layer.

13.2 Caching Strategy

- Static pages (landing, login): Next.js SSG — no server work per request
- Dashboard data: SWR (stale-while-revalidate) — show cached data instantly, refresh in background
- AI-generated plans for common exams: Cached in Supabase ai_plan_cache table, TTL 30 days
- Charts data: Memoized with useMemo, only recomputes when date range changes

14. Development Roadmap

Phase	Duration	Deliverables	Priority
Phase 0 — Foundation	1 week	Next.js setup, Supabase config, auth, CI/CD, design system tokens, shadcn/ui base components	Must-have
Phase 1 — Planner Core	2 weeks	Plan CRUD, topic/subtopic management, manual scheduling algorithm, basic plan detail view	Must-have
Phase 2 — Habit Tracker	1.5 weeks	Habit CRUD, daily log, streak calculation, habit list view, week strip	Must-have
Phase 3 — Dashboard	1 week	Dashboard layout, stat cards, today's focus, habit rings, streak badge with animations	Must-have
Phase 4 — Analytics	1 week	Heatmap, charts (Recharts), progress page, streak history, completion rates	Must-have
Phase 5 — Animations	1 week	GSAP scroll triggers, Framer Motion page transitions, streak animations, Lottie icons	High
Phase 6 — AI Planner	1 week	AI API integration, editable review table, PDF parsing, caching common plans	High
Phase 7 — Notifications	3 days	Web Push, Cron jobs, in-app notification center, streak risk alerts	Medium
Phase 8 — Polish	1 week	Mobile responsiveness, performance optimization, edge cases, error states, empty states	High
Phase 9 — Deployment	2 days	Vercel deploy, env vars, domain, monitoring (Vercel Analytics), Sentry error tracking	Must-have

14.1 Total Estimated Timeline: ~10–11 weeks

This assumes a single developer working approximately 6-8 hours per day. Phases can be parallelized with a team.

15. Open Source & Free Tier Summary

Service	Free Tier Limits	Upgrade Cost	Mitigation
Supabase	500MB DB, 50K MAU, 5GB bandwidth	$25/mo Pro	Compress data, lazy-load history
Vercel	Hobby: 100GB bandwidth, 100 deploys/day	$20/mo Pro	Sufficient for personal/small project
NextAuth.js	Unlimited, self-hosted	Free forever	None needed
GSAP	Free for personal/open-source	Club for commercial	Keep project open-source
Framer Motion	MIT, completely free	Free forever	None needed
Lucide React	MIT, completely free	Free forever	None needed
Lottie Files	Free animations available	$39/mo for Pro animations	Use free Lottie animations only
Recharts	MIT, completely free	Free forever	None needed
Claude API	No free tier	~$3-15 per 1M tokens	Cache common plans, rate-limit AI calls
Drizzle ORM	MIT, completely free	Free forever	None needed

Total Monthly Cost for Personal Use

Supabase Free + Vercel Hobby + Claude API (~$2-5/month for moderate AI usage) = ~$2-5/month total.For a team or production app with heavy usage, budget ~$50-70/month for Supabase Pro + Vercel Pro + Claude API.

16. Recommended Project Structure

```
studyflow-pro/├── app/                          # Next.js 14 App Router│   ├── (auth)/login/
# Auth pages (public)│   ├── (dashboard)/              # Protected layout group│   │   ├──
dashboard/page.tsx│   │   ├── planner/│   │   ├── page.tsx          # Plan list│   │   │
├── new/page.tsx      # Create plan│   │   │   └── [id]/page.tsx     # Plan detail│   │   ├──
habits/│   │   │   ├── page.tsx│   │   │   └── progress/page.tsx│   │   └── analytics/page.tsx│
├── api/                      # API routes (server-side)│   │   ├── auth/[...nextauth]/│   │
├── plans/│   │   ├── topics/│   │   ├── habits/│   │   ├── progress/│   │   ├── streaks/│   │
├── analytics/│   │   └── ai/generate-plan/│   └── layout.tsx├── components/│   ├── ui/
# shadcn/ui base components│   ├── dashboard/                # Dashboard-specific components│
├── planner/                  # Planner components│   ├── habits/                  # Habit
tracker components│   ├── charts/                  # Recharts wrappers│   ├── animations/
# Reusable animation components│   └── common/                  # Shared (Navbar, Sidebar,
etc.)├── lib/│   ├── db/                      # Drizzle schema + client│   │   ├── schema.ts│
│   └── index.ts│   ├── auth.ts                    # NextAuth config│   ├── supabase.ts
# Supabase client│   ├── scheduler.ts              # Plan scheduling algorithm│   ├── streak.ts
# Streak calculation logic│   └── ai.ts                    # AI API integration├── store/
# Zustand stores│   ├── planStore.ts│   ├── habitStore.ts│   └── uiStore.ts├── hooks/
# Custom React hooks├── types/                       # TypeScript types├── drizzle.config.ts├──
tailwind.config.ts└── next.config.ts
```

Appendix: Design Reference Analysis

Image Ref	App/Design	Key Takeaway for StudyFlow Pro
Image 1	Routiner — Habit Tracker mobile	Clean habit list UI, category chips, progress indicators, blue/white mobile-first design
Image 2	Virtus — Dark Task Dashboard	Dark card grid layout, orange accent, performance graph, task progress %, team assignment widgets
Image 3	Everyday — Habit Heatmap	Year-view heatmap with color intensity, streak counts, habit rows with cross-habit calendar view
Image 4/5	HabitKit — Stats Cards	Dot grid for habit history, streak timeline bars, completion rate card, floating card layout
Image 6	HabitBull — Mobile habit app	Streak stats, habit history bar chart, monthly calendar heatmap in pink/magenta color scheme
Image 7/9	Habit Tracker app	Yes/No/Skip action buttons, weekly strip view, pie chart for done/missed, time range toggle
Image 8	Notion Habit Tracker	Minimalist entry-based design, habit list as database, calendar navigation, earthy tones
Image 10	Tasker — Task Management	Gantt-style timeline view, progress cards with team avatars, stat count widgets, purple gradient dark UI
Image 11	Dark Mode HR App	Responsive mobile dark screens, recruitment/headcount UI, proves dark mode works on mobile
Image 12	Dark Team Calendar	Multi-team calendar with event cards, right-side detail panel, week view navigation, dark purple gradient

— End of Document —

StudyFlow Pro PRD v1.0 | February 2026